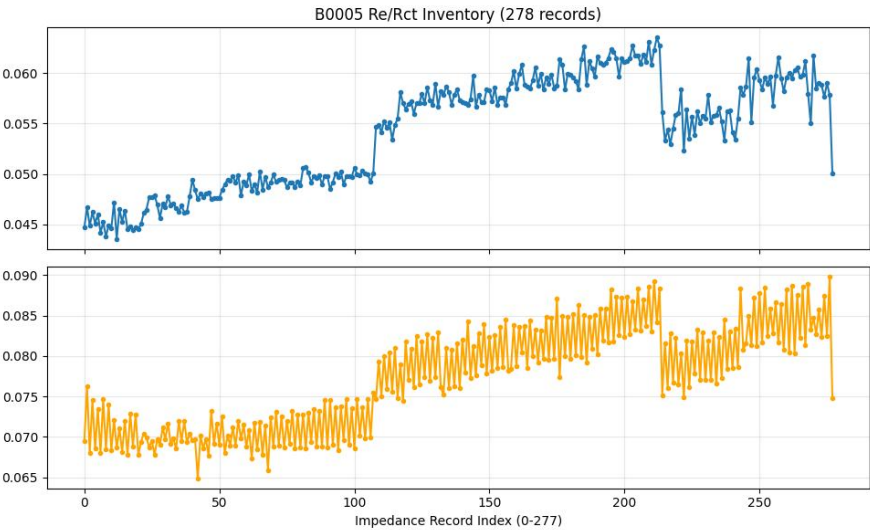


CASE-001 — WHEN SOH LOOKS WRONG

Status: Completed — DS-001 NASA B0005 — 168 discharges / 278 impedance / 5 regenerations

TECHNICAL QUESTION: Can a single capacity measurement be trusted as a direct indicator of battery degradation?

EVIDENCE 01-03 VISUAL: Re/Rct NON-MONOTONIC



Capacity: 1.8565 → 1.3251 (-28.6%), 5 upward excursions, max +0.0913 at idx 89
Impedance: Re 0.0436→0.0635 (+45% range), Rct 0.0649→0.0898 (+38% range), both show step drops at idx ~214
Time: impedance intervals 1:20:28 to 3d 2:46:18, median ~3.34h — highly non-uniform

CASE-001 METHOD: Before deciding what is wrong, determine whether the data supports the conclusion.

FOUR EVIDENCES — WHAT THE DATA SUPPORTS

E01 CAPACITY	E02 TIME GAPS	E03 IMPEDANCE	E04 AMBIENT T
168 discharge records. Overall decline but not monotonic. 5 prominent regenerations at D18/30/89/149/166.	Prev D→Regen: 4h to >1d. Prev Imp→Regen: 0:31:57 to 15:24:46, 1 with no preceding Imp. No single consistent pattern.	278 records. 277 intervals >0. Re +12.02% first→last, Rct +7.68% but both non-monotonic with sharp drops.	24.0°C at all 5 events. No ambient variation observed. Ambient ruled out as explanation in this dataset.

ENGINEERING JUDGMENT

A measured capacity value is not equivalent to underlying irreversible degradation at that moment. Short-term increases are consistent with relaxation / measurement history effects, observed within minutes (31:57 gaps) in a 24°C controlled chamber. This case does not prove a single electrochemical mechanism.

HEALTH = TREND + CONTEXT, NOT SINGLE ABSOLUTE

BESS RULE: Do not interpret every up/down SOH point as recovery/degradation. Evaluate trend + context: time since last full cycle, rest duration, diagnostic history, temp, operating condition, persistence. A system treating every point as truth will generate false alarms.

evidence/: capacity_fade_final.png / Evidence-02-Time-Gaps.md / B0005_Re_Rct_Inventory_278_records.png / impedance_inventory.csv